Order of Operations with Integers (GEMDAS)

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Numerical Expressions

Numerical expressions are mathematical phrases involving numbers and operations such as addition, subtraction, multiplication, division, and exponentiation.

$$5 + 2(3 - 6)$$

This is a numerical expression.

Number properties help us simplify numerical expressions.

Commutative Property

Changing the **order** of the numbers does not change the result of addition or multiplication.

$$(-3) + 5 = 5 + (-3)$$

$$(-2)(4) = (4)(-2)$$

Associative Property

Changing the **grouping** of numbers does not change the result of addition or multiplication.

$$[(-2) + 4] + 6 = (-2) + [4 + 6]$$

Both sides equal 8.

Distributive Property

Multiplication can be **distributed** over addition or subtraction.

$$3(4+2) = 3(4) + 3(2)$$

Both sides equal 18.

Identity Property

Adding zero or multiplying by one leaves a number unchanged.

$$-7 + 0 = -7$$

$$(-7)(1) = -7$$

Additive Inverse Property

A number added to its opposite equals zero.

$$(-8) + 8 = 0$$

Order of Operations

Order of Operations (GEMDAS) tells us which operations to perform first when simplifying numerical expressions.

Without a rule, $2 + 3 \times 4$ could give two different answers!

GEMDAS

G	Grouping Symbols
E	Exponents
MD	Multiplication and Division
AS	Addition and Subtraction

Work from top to bottom.

Memory Aid

Great **É**lephants **M**ake **D**elicious **A**pple **S**moothies

Grouping, **E**xponents, **M**ultiplication and **D**ivision, **A**ddition and **S**ubtraction

G – Grouping Symbols

Simplify expressions inside parentheses, brackets, and braces first.

$$(\quad) \qquad [\quad] \qquad \{ \quad \}$$

$$4(3+2) = 4(5) = 20$$

E – Exponents

Evaluate powers and exponents.

$$(-3)^2 = (-3)(-3) = 9$$

$$2^3 = 2 \times 2 \times 2 = 8$$

MD – Multiplication and Division

Perform multiplication and division from **left to right**.

$$\begin{aligned} 18 \div 3 \times 2 &= 6 \times 2 \\ &= 12 \end{aligned}$$

AS – Addition and Subtraction

Perform addition and subtraction from **left to right**.

$$\begin{aligned} 10 - 4 + 3 &= 6 + 3 \\ &= 9 \end{aligned}$$

Important Reminder

Multiplication does not always come before division, and addition does not always come before subtraction.

Operations at the same level of priority are performed from **left to right**.

Same Level: Left to Right

$$12 \div 3 \times 2 = 4 \times 2 = 8$$

Not $12 \div 6 = 2$.

$$10 - 4 + 3 = 6 + 3 = 9$$

Not $10 - 7 = 3$.

Golden Rule

Follow GEMDAS carefully, simplify one step at a time, and pay attention to the signs of integers.

Examples

Example 1

Simplify.

$$\begin{aligned} & 5 + 2(3 - 6) \\ & = 5 + 2(-3) \\ & = 5 + (-6) \\ & = -1 \end{aligned}$$

Example 2

Simplify.

$$
\begin{aligned}
&(-2)^2 + 18 \div (-3) \\
&= 4 + 18 \div (-3) \\
&= 4 + (-6) \\
&= -2
\end{aligned}
$$

Example 3

Simplify.

$$\begin{aligned} & 24 \div (-6) \times 2 \\ &= (-4) \times 2 \\ &= -8 \end{aligned}$$

Division comes first because it is the leftmost.

Example 4

Simplify.

$$
\begin{aligned}
&2\{5+(-3)^2\}-8 \\
&=2\{5+9\}-8 \\
&=2\{14\}-8 \\
&=28-8 \\
&=20
\end{aligned}
$$

Practice Exercises

Practice 1

Simplify the numerical expression.

$$(-8) + 3(4 - 7)$$

Practice 2

Simplify the numerical expression.

$$24 \div (-6) + 5(-2)$$

Practice 3

Simplify the numerical expression.

$$(-3)^2 + 4(-5)$$

Practice 4

Simplify the numerical expression.

$$18 - [6 + 2(-4)]$$

Practice 5

Simplify the numerical expression.

$$(-2)[5 + (-3)]^2$$

Practice 6

Simplify the numerical expression.

$$36 \div [(-3)(-2)] - 7$$

Practice 7

Simplify the numerical expression.

$$4\{3 + (-2)^3\} - 5$$

Practice 8

Simplify the numerical expression.

$$(-15) \div 3 + 2[4 - (-6)]$$

Practice 9

Simplify the numerical expression.

$$5[(-4)+7]-2^3$$

Practice 10

Simplify the numerical expression.

$$30 - \{18 \div (-3) + 2[(-5) + 8]\}$$

Activities

Activity 1

Simplify the numerical expression.

$$(-6) + 4(2 - 5)$$

Activity 2

Simplify the numerical expression.

$$35 \div (-7) + 3(-4)$$

Activity 3

Simplify the numerical expression.

$$(-4)^2 + 3(-6)$$

Activity 4

Simplify the numerical expression.

$$20 - [8 + 3(-2)]$$

Activity 5

Simplify the numerical expression.

$$(-3)[4 + (-2)]^2$$

Activity 6

Simplify the numerical expression.

$$48 \div [(-4)(-3)] - 6$$

Activity 7

Simplify the numerical expression.

$$5\{2 + (-2)^3\} - 4$$

Activity 8

Simplify the numerical expression.

$$(-24) \div 4 + 3[5 - (-7)]$$

Activity 9

Simplify the numerical expression.

$$6[(-5)+8]-3^2$$

Activity 10

Simplify the numerical expression.

$$40 - \{24 \div (-4) + 3[(-6) + 9]\}$$

Thank you!